

CSA5112 - Cryptography and Network Security

5. Playfair Cipher

Aim: To implement the Playfair Cipher for encrypting a pair of characters using a 5×5 matrix.

Algorithm:

1. Start.
2. Create a 5×5 key matrix.
3. Divide plaintext into pairs.
4. Apply Playfair encryption rules:
 - Same row $\rightarrow$ shift right.
 - Same column $\rightarrow$ shift down.
 - Rectangle $\rightarrow$ exchange columns.
5. Display ciphertext.
6. Stop.

Code:

```
#include<stdio.h>

#include<string.h>

#include<ctype.h>

char matrix[5][5];

void generateMatrix(char key[])

{

    int used[26]={0};

    int i,j,k=0,row=0,col=0;

    used['J'-'A']=1;

    for(i=0;key[i]!='\0';i++)

    {
```

```
char ch=toupper(key[i]);

if(ch=='J')

    ch='I';

if(ch<'A' || ch>'Z')

    continue;

if(!used[ch-'A'])

{

    used[ch-'A']=1;

    matrix[row][col]=ch;

    col++;

    if(col==5)

    {

        row++;

        col=0;

    }

}

for(i=0;i<26;i++)

{

    if(!used[i])

    {

        matrix[row][col]=i+'A';

        col++;

        if(col==5)
```

```

        {
            row++;

            col=0;
        }
    }

}

}

void findPosition(char ch,int *r,int *c)

{
    if(ch=='J')

        ch='I';

    int i,j;

    for(i=0;i<5;i++)

        for(j=0;j<5;j++)

            if(matrix[i][j]==ch)

                {

                    *r=i;

                    *c=j;

                    return;

                }

}

int main()

{

    char key[100],text[100];

```

```
int i,r1,c1,r2,c2;

printf("Enter Key: ");

scanf("%s",key);

printf("Enter Plain Text (even letters): ");

scanf("%s",text);

generateMatrix(key);

printf("\nKey Matrix:\n");

for(i=0;i<5;i++)

{

    int j;

    for(j=0;j<5;j++)

        printf("%c ",matrix[i][j]);

    printf("\n");

}

printf("\nCipher Text: ");

for(i=0;text[i]!='\0';i+=2)

{

    findPosition(toupper(text[i]),&r1,&c1);

    findPosition(toupper(text[i+1]),&r2,&c2);

    if(r1==r2)

    {

        printf("%c%c",matrix[r1][(c1+1)%5],matrix[r2][(c2+1)%5]);

    }

    else if(c1==c2)
```

```

    {
        printf("%c%c",matrix[(r1+1)%5][c1],matrix[(r2+1)%5][c2]);
    }
    else
    {
        printf("%c%c",matrix[r1][c2],matrix[r2][c1]);
    }
}
return 0;
}

```

Sample Input:

```

Enter Key: MONARCHY
Enter Plain Text: HI

```

Sample Output:

```

Key Matrix:
M O N A R
C H Y B D
E F G I K
L P Q S T
U V W X Z

Cipher Text: BF

```

Result: The Playfair Cipher algorithm was successfully implemented using a 5×5 key matrix, and the plaintext was encrypted according to Playfair encryption rules.